



Figure 1: HumanPulse Protocol Logo

## HumanPulse Protocol — Business Plan v1

**Project:** HumanPulse Protocol (\$HPP)

**Network:** Solana

**Stage:** Hackathon MVP / early protocol validation

**Document type:** Business plan and go-to-market roadmap

**Primary technical source of truth:** HumanPulse Protocol Technical Whitepaper v7.2

**Positioning statement:**

**HumanPulse proves the presence of a unique living human at a specific moment in time — instantly, privately, and on-chain.**

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### 1. Executive Summary

HumanPulse Protocol is a privacy-first Proof of Unique Humanity infrastructure for Solana.

The protocol is designed to solve a growing problem across crypto and digital networks: applications increasingly need to distinguish unique living humans from bots, duplicate wallets, Sybil farms, and automated agents, without forcing users into centralized identity databases or long, invasive verification flows.

The current hackathon MVP demonstrates a focused version of this vision:

1. a browser-based webcam challenge;
2. local MediaPipe hand detection;
3. wallet connection on Solana Devnet;
4. a real proof-consumption transaction on Solana Devnet;
5. a deployed \$HPP Devnet token;
6. a deployed Anchor burn-on-use program representing the intended protocol fee mechanism;
7. a live demo website and public repository package.

The long-term business opportunity is to become a verification infrastructure layer for applications that need Sybil resistance: DAOs, launchpads, airdrops, token launches, DePIN networks, governance platforms, wallets, exchanges, and gated digital applications.

HumanPulse does not aim to sell identity. It aims to sell **proof-consumption events**: privacy-preserving verification moments that applications can consume when they need stronger confidence that an action is being performed by a unique living human.

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### 2. The Problem

Digital systems are entering an era where artificial agents, bot farms, wallet farms, synthetic identities, and deepfake-capable systems can operate at scale.

For crypto networks, this creates direct economic damage:

- airdrops can be farmed by thousands of duplicated wallets;
- token launches can be manipulated by scripted participation;
- DAO voting can be distorted by Sybil wallets;
- DePIN rewards can be extracted by fake or duplicated participants;
- gated access can be bypassed by automated accounts;
- reputation systems can be polluted by non-human or duplicated actors.

The key problem is not simply “identity.” The key problem is **unique human presence at the moment of action.**

Most existing approaches create trade-offs:

- CAPTCHAs are increasingly weak against automation.
- Centralized KYC is high-friction and privacy-invasive.
- Social graph methods can be gamed or are slow to bootstrap.
- Wallet reputation does not prove a live human is present.
- Biometric databases introduce custody risk and regulatory sensitivity.
- Proof-of-personhood systems can become heavy, slow, or socially complex.

HumanPulse addresses a narrower and more composable primitive:

Prove that a unique living human is present at a specific moment in time, without storing biometric identity.

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### 3. Product Overview

HumanPulse is designed as a verification infrastructure layer that applications can call when they need a privacy-preserving human-presence proof.

The intended production flow is:

1. A user attempts a high-value action, such as claiming an airdrop, voting, joining a gated community, or receiving DePIN rewards.
2. The application requests a HumanPulse verification.
3. The user completes a liveness/challenge flow.
4. HumanPulse generates or consumes a proof.
5. The application receives a proof-consumption signal.
6. The action is allowed, limited, weighted, or rejected based on protocol rules.

The current MVP is deliberately narrower:

- webcam-based challenge-response using local browser hand detection;
- proof-consumption event written to Solana Devnet;
- no storage of webcam frames;
- no biometric database;
- no production HumanPulse validator network yet;
- no production zk-STARK proof pipeline yet.

This distinction is important. The MVP proves that HumanPulse can deliver a real Solana-native verification flow today, while the whitepaper and roadmap describe how the protocol can evolve into a broader Proof of Unique Humanity infrastructure layer.

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## 4. Current MVP vs Technical Roadmap

### 4.1 Current MVP / Hackathon Implementation

The current MVP includes:

- live demo website;

- public landing/demo repository;
- browser-based MediaPipe hand-detection challenge;
- local challenge-response flow;
- wallet connection through Phantom on Solana Devnet;
- real Solana Devnet proof-consumption transaction;
- \$HPP Devnet token;
- deployed Anchor burn-on-use program;
- public technical whitepaper v7.2;
- pitch deck v9;
- README documentation clarifying MVP vs roadmap.

In the current hackathon MVP, the browser acts as the local challenge-response verifier. After the challenge is completed, the user signs a Solana Devnet transaction that records a proof-consumption event on-chain.

Solana Devnet validates the transaction. The full HumanPulse validator network is not yet live in the MVP.

## 4.2 Technical Roadmap

The technical roadmap includes:

- HumanPulse validator / aggregator network;
- recursive zk-STARK aggregation;
- TEE-based local proof generation;
- production-grade burn-on-use \$HPP verification fee flow;
- HPP-first gas abstraction;
- relayer / fee-payer infrastructure;
- validator reward mechanisms;
- optional advanced sensing modules.

## 4.3 Optional Research Modules

Optional research modules include:

- LiDAR photonics;
- pupillary reflex;
- micro-expression analysis;
- multi-device verification;
- advanced liveness fusion;
- additional anti-spoofing signals.

These modules are not required for the current MVP and should be treated as research roadmap, not as hackathon implementation claims.

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## 5. Target Customers

HumanPulse is primarily a B2B / protocol infrastructure product. The strongest early customers are applications that suffer direct economic loss from Sybil behavior.

### 5.1 DAOs and Governance Platforms

DAOs need to reduce vote manipulation, duplicated participation, and governance attacks. HumanPulse can be used before voting, delegation, proposal creation, or high-impact governance actions.

## 5.2 Airdrop Platforms

Airdrops are one of the strongest initial use cases. Projects want to distribute tokens to real users, not wallet farms. HumanPulse can act as a pre-claim or claim-weighting verification layer.

## 5.3 Token Launches and Launchpads

Token launches are vulnerable to botting, allocation farming, and scripted wallet participation. HumanPulse can be used to gate access, limit duplicate participation, or improve fairness.

## 5.4 DePIN Networks

DePIN reward systems need stronger guarantees that users, operators, or devices represent legitimate participation. HumanPulse can support human-bound verification for reward claims and user onboarding.

## 5.5 Wallets, Exchanges, and Consumer Apps

Wallets and exchanges may use HumanPulse to protect specific high-risk actions without forcing full KYC on every user interaction.

## 5.6 Gated Communities and Digital Access

HumanPulse can support privacy-preserving access gates where the requirement is not legal identity, but live human presence.

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# 6. Core Use Cases

## 6.1 Sybil-Resistant Airdrop Claims

A project can require HumanPulse verification before allowing an airdrop claim. The application consumes a HumanPulse proof and allows the wallet to proceed.

Business value:

- fewer duplicate claims;
- fairer distribution;
- reduced bot extraction;
- better community quality.

## 6.2 DAO Voting Protection

A DAO can require HumanPulse verification before a vote, or use it as a weighting input.

Business value:

- stronger governance legitimacy;
- reduced fake participation;
- higher confidence in human voter presence.

## 6.3 Token Launch Fairness

A launchpad can require verification before allocation.

Business value:

- reduced wallet farming;
- improved fairness;

- better launch credibility.

#### **6.4 DePIN Reward Eligibility**

A DePIN network can require periodic proof-of-humanity checks for certain user reward flows.

Business value:

- reduced fake accounts;
- improved reward efficiency;
- better network integrity.

#### **6.5 Gated Access**

A digital platform can use HumanPulse to allow one-human-one-access flows without collecting legal identity.

Business value:

- lower friction than KYC;
  - stronger anti-bot control than CAPTCHA;
  - better privacy posture.
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### **7. Revenue Model**

HumanPulse monetizes verification events.

The protocol is designed around a usage-driven model:

Applications or users consume \$HPP when they request a HumanPulse verification.

This creates a direct link between protocol usage and token utility.

Potential pricing models:

#### **7.1 Per-Verification Fee**

Applications pay a fixed or dynamic fee per verification event.

Example:

- \$HPP fee per airdrop claim;
- \$HPP fee per governance vote;
- \$HPP fee per token launch registration;
- \$HPP fee per gated access event.

#### **7.2 Application-Funded Verification Pools**

A DAO, launchpad, or DePIN application pre-funds a pool of HumanPulse verifications.

The user experience can be free for the user while the application pays for Sybil resistance.

This is likely the strongest B2B model.

#### **7.3 Tiered Enterprise / Protocol Access**

Larger integrations may pay for volume packages, dedicated API support, analytics, and higher reliability.

Example tiers:

- Startup / community tier;
- DAO / launchpad tier;
- enterprise / infrastructure tier.

#### **7.4 Protocol-Level Verification Markets**

In a more mature network, third-party relayers, validators, and proof aggregators can participate in the verification economy, receiving protocol rewards for useful infrastructure work.

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### **8. \$HPP Token Utility**

\$HPP is intended to be the economic token of the HumanPulse protocol.

Its core utilities are:

1. verification consumption;
2. burn-on-use mechanics;
3. validator / aggregator rewards;
4. relayer incentives;
5. gas abstraction reserve funding;
6. protocol access and integration economics.

The key design principle is:

SOL should not be the economic token of HumanPulse. SOL is only the underlying Solana network fee. \$HPP is the protocol usage token.

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### **9. HPP-First Gas Abstraction Model**

A major UX requirement for global adoption is that end users should not need to manually acquire SOL just to complete a HumanPulse verification.

On Solana, transactions require SOL for network fees. HumanPulse cannot remove this network-level requirement. However, the protocol can abstract it from the end user.

The intended production model is:

1. the user or application consumes \$HPP for a verification;
2. the app, relayer, or protocol fee-payer wallet pays the SOL network fee;
3. a portion of \$HPP verification fees funds a Gas Abstraction Reserve;
4. the reserve maintains SOL liquidity for fee-payer wallets;
5. the end-user experience remains HPP-first or even fully sponsored by the application.

#### **9.1 Example Fee Split**

A future verification fee could be split as follows:

- 50% burn;
- 30% validators / aggregators;
- 20% Gas Abstraction Reserve / relayers.

This split is illustrative and can be adjusted by governance or treasury policy.

#### **9.2 Why This Matters**

If users had to manually buy SOL just to verify, the UX would become weaker and \$HPP would appear less central.

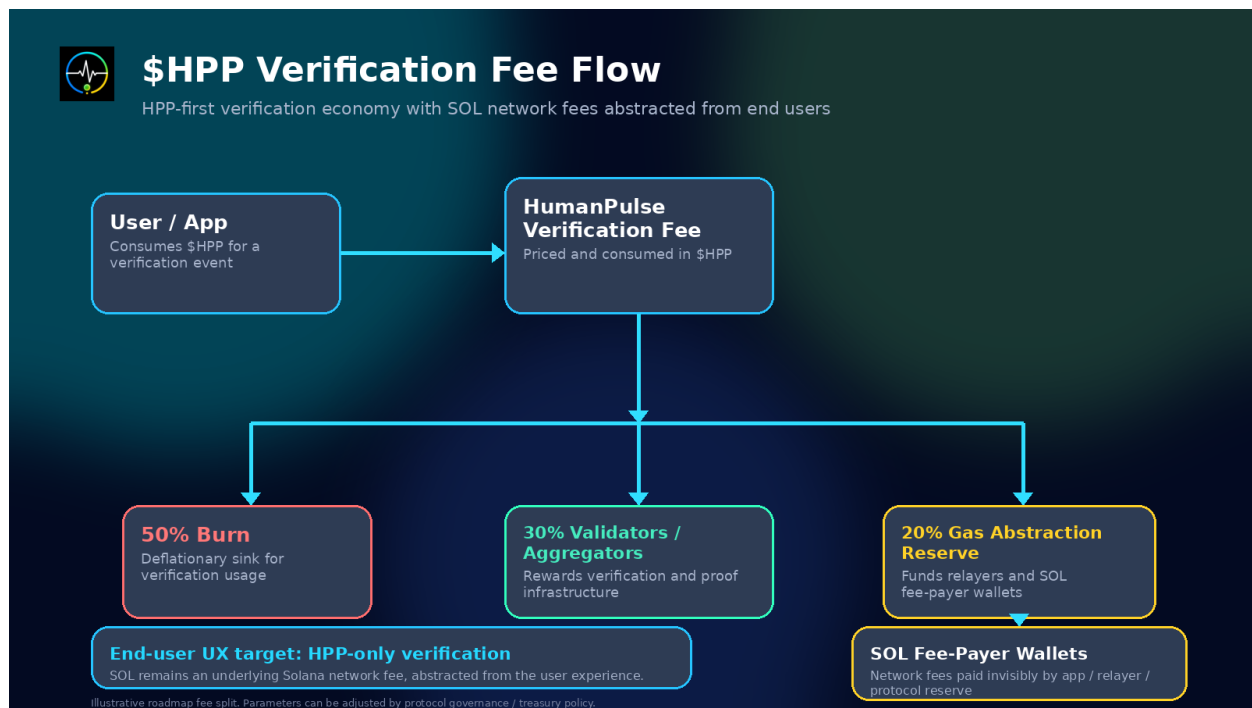


Figure 2: HPP Verification Fee Flow

The HPP-first model keeps \$HPP as the economic layer:

- users or apps consume \$HPP;
- the protocol handles SOL behind the scenes;
- relayers or apps sponsor network fees;
- gas abstraction makes onboarding easier;
- \$HPP remains tied to real verification demand.

### 9.3 Operational Treasury Model

At early stages, this can be managed semi-automatically:

- the protocol treasury or multisig holds SOL in fee-payer wallets;
- a portion of \$HPP fees accumulates in the Gas Abstraction Reserve;
- treasury operations periodically rebalance HPP/SOL;
- fee-payer wallets are topped up based on verification volume.

At later stages, this can evolve into:

- automated treasury policies;
- price-aware rebalancing;
- relayer marketplaces;
- application-level fee delegation;
- validator / relayer compensation in \$HPP.

## 10. Unit Economics

A verification event should be priced to cover:

1. protocol margin;
2. validator / aggregator compensation;
3. burn mechanics;

4. relay / gas abstraction costs;
5. operational risk buffer.

A simplified formula:

HPP verification fee =  
base protocol fee  
+ validator / aggregator reward  
+ gas abstraction contribution  
+ volatility / retry buffer

Because SOL and \$HPP have different market values, the Gas Abstraction Reserve should not attempt to cover gas on a one-to-one fixed basis. Instead, it should maintain a target SOL balance based on expected verification volume.

Example:

Target SOL balance = projected network fees for N future verifications × safety buffer

This allows the protocol to support a gasless or HPP-only user experience while protecting itself from volatility, congestion, retries, and liquidity changes.

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## **11. Go-To-Market Strategy**

HumanPulse should begin with high-pain crypto-native use cases where the value of Sybil resistance is immediate and easy to understand.

### **11.1 Phase 1 — Solana Hackathon and Developer Credibility**

Goals:

- demonstrate a live working MVP;
- publish public repositories;
- show Devnet proof-consumption transactions;
- present a clear whitepaper, pitch deck, and business model;
- attract developer, judge, and ecosystem attention.

Primary audience:

- Solana judges;
- developer community;
- early DAO and launchpad operators;
- identity and anti-Sybil infrastructure builders.

### **11.2 Phase 2 — Airdrop and Launchpad Pilots**

The first commercial pilots should target:

- airdrop platforms;
- token launch platforms;
- NFT allowlists;
- DAO claim gates;
- community onboarding flows.

Reason:

- Sybil abuse is obvious;
- ROI is easy to explain;
- verification can be inserted before a claim or registration;
- integration surface is simple.



### 11.3 Phase 3 — DAO Governance and DePIN

After initial pilots, HumanPulse can expand into:

- DAO voting;
- proposal creation;
- gated governance forums;
- DePIN reward eligibility;
- periodic human-presence checks.

### 11.4 Phase 4 — Wallet and Consumer Distribution

Longer-term distribution can come through wallets, dApps, and consumer platforms that integrate HumanPulse as a reusable verification primitive.

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## 12. Market Positioning

HumanPulse should not position itself as another CAPTCHA.

It should position itself as:

a privacy-first human-presence proof layer for high-value on-chain actions.

The strongest comparison points are:

- CAPTCHA: low assurance, web-native, weak against automation.
- KYC: high assurance, high friction, privacy-sensitive.
- Social graph proof: useful but slow and gameable.
- Wallet reputation: does not prove live human presence.
- HumanPulse: instant, private, on-chain proof consumption for live human presence.

HumanPulse's wedge is not legal identity. It is **proof of unique live human presence at the moment of action**.

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## 13. Competitive Advantage

HumanPulse can build advantage through:

1. Solana-native speed and composability;
2. privacy-first architecture;
3. no biometric database positioning;
4. usage-based \$HPP economy;
5. gas abstraction roadmap;
6. verifier / aggregator roadmap;
7. strong first use case in Sybil resistance;
8. modular architecture for DAOs, launches, DePIN, and gated access.

The protocol should be careful not to overclaim. The correct positioning is:

- designed to minimize identity custody;
  - intended to increase Sybil attack cost;
  - current MVP demonstrates local challenge-response and Devnet proof consumption;
  - validator network and zk aggregation are roadmap components.
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## 14. Key Metrics

HumanPulse should track metrics that connect technical use to business value.

### Product Metrics

- number of verification attempts;
- completion rate;
- challenge failure rate;
- average verification time;
- wallet connection success rate;
- transaction confirmation success rate.

### Protocol Metrics

- number of proof-consumption events;
- \$HPP consumed;
- \$HPP burned;
- validator / relayer rewards;
- Gas Abstraction Reserve balance;
- SOL fee-payer wallet runway.

### Business Metrics

- number of integrations;
- verifications per integration;
- cost per verification;
- revenue per verification;
- conversion rate from demo to pilot;
- pilot-to-production conversion.

### Trust Metrics

- privacy incidents;
  - false positive / false negative rates;
  - spoofing resistance improvements;
  - audit status;
  - validator uptime when validator network launches.
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## 15. Risks and Mitigations

### 15.1 Technical Risk

Risk: browser-based verification can be spoofed or fail under poor camera conditions.

Mitigation:

- treat MVP as challenge-response demo;
- roadmap toward stronger liveness signals;
- add validator / aggregator layer;
- introduce zk and TEE-based proof generation.

### 15.2 Overclaim Risk

Risk: users or judges may think the MVP already includes full biometric proof generation or validator consensus.

Mitigation:

- clearly separate MVP from roadmap;
- document that current MVP uses local MediaPipe hand detection;
- state that validator network and zk aggregation are roadmap components.

### **15.3 Token UX Risk**

Risk: requiring SOL for verification weakens the \$HPP experience.

Mitigation:

- HPP-first gas abstraction roadmap;
- sponsored transactions;
- relayers;
- protocol-managed SOL fee-payer wallets.

### **15.4 Regulatory / Privacy Risk**

Risk: biometric systems may raise privacy and compliance concerns.

Mitigation:

- avoid storing raw biometric data;
- process camera frames locally in MVP;
- design protocol around proof consumption, not identity custody;
- avoid legal identity claims.

### **15.5 Adoption Risk**

Risk: applications may not integrate a new verification layer.

Mitigation:

- start with high-pain use cases;
  - provide simple widget / API;
  - demonstrate clear ROI for airdrops and launches;
  - build Solana ecosystem partnerships.
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## **16. 12-Month Roadmap**

### **0-3 Months**

- finalize hackathon submission;
- stabilize live demo;
- publish public documentation;
- approach early Solana DAOs and launchpads;
- begin Anchor burn-on-use frontend integration;
- define validator / relayer economics.

### **3-6 Months**

- build first pilot integrations;
- implement HPP-based verification consumption;
- introduce sponsored transaction / fee-payer prototype;
- improve challenge-response robustness;
- design validator onboarding model.

## 6-9 Months

- launch private beta with selected partners;
- add richer proof-consumption analytics;
- test Gas Abstraction Reserve operations;
- begin security review of smart contracts;
- design zk aggregation prototype scope.

## 9-12 Months

- public beta;
  - validator / aggregator testnet;
  - relayer fee sponsorship model;
  - partner integrations;
  - publish updated protocol economics paper;
  - prepare production-grade audit roadmap.
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## 17. Business Conclusion

HumanPulse Protocol has a clear business wedge:

high-value applications need Sybil resistance, but users do not want centralized identity custody.

The protocol can monetize verification events while preserving a privacy-first architecture. \$HPP is designed to become the economic layer of verification consumption, while SOL network fees can be abstracted from the user through relayers, sponsored transactions, and a protocol-managed Gas Abstraction Reserve.

The current MVP is intentionally narrow but real: it demonstrates browser-based challenge-response and a live Solana Devnet proof-consumption transaction.

The broader opportunity is to turn HumanPulse into a Solana-native human-presence infrastructure layer for the AI era.

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## 18. Official Links

- Main Protocol Repository: <https://github.com/humanpulseprotocol/HumanPulse-Protocol>
  - Landing Demo Repository: <https://github.com/humanpulseprotocol/humanpulse-landing>
  - Live Website: <https://humanpulse-landing.vercel.app/>
  - Demo Video: <https://humanpulse-landing.vercel.app/HumanPulse-demo.mp4>
  - Solana Devnet Program: <https://explorer.solana.com/address/D9tRQi8nZzARTJZXTmSsy4hrx4BwKUNPezKza>
  - \$HPP Token on Solana Devnet: <https://explorer.solana.com/address/EsBZKfFmhVAypmVm6QmenevZ56XZQ2>
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## 19. Final Note

This document describes the business and go-to-market roadmap for HumanPulse Protocol.

It should be read together with:

- HumanPulse Technical Whitepaper v7.2;
- Solana Frontier Pitch Deck v9;
- public GitHub repositories;
- live Devnet demo.

The business model, fee splits, gas abstraction reserve, and validator economics are roadmap design elements and may evolve through testing, governance, and production implementation.